

Two cases of the A_2 Andrews–Gordon conjectures for cylindric partitions ($k = 3$ at modulus 11, $k = 4$ at modulus 13), and a bounded refinement of positivity at $d = 4$

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Abstract

Building on results of Corteel–Welsh, Kanade–Russell and Uncu, we prove the first two cases beyond $k = 2$ of Warnaar’s fermionic-form conjectures for cylindric partitions [7, Conjectures 2.5 and 2.11]: the balanced profiles $c = (3, 3, 2)$ at $d = 8$ (modulus 11, $k = 3$) and $c = (4, 3, 3)$ at $d = 10$ (modulus 13, $k = 4$). Both proofs chain Warnaar’s own finite form [7, Proposition 7.1], in a coefficientwise limit, with an exact Pochhammer split and one of Uncu’s theorems. At modulus 11 one Kanade–Russell contiguous relation [4, Lemma 9.2] is also needed to reach Uncu’s modulus-11 theorem [6, Theorem 4.3]; at modulus 13 the chain is *shorter* — the split lands verbatim on Uncu’s modulus-13 theorem [6, Theorem 5.3], and no contiguous relation is required. As a corollary, the positivity conjecture [7, Conjecture 2.7] holds for the orbits of $(3, 3, 2)$ and $(4, 3, 3)$. We also record a bounded refinement at $d = 4$: for every profile c with $c_0 + c_1 + c_2 = 4$ one has $H_{c,m} = \sum_{n \leq m} \begin{bmatrix} m \\ n \end{bmatrix} Q_{n,c}$ with the $Q_{n,c}$ given by explicit manifestly nonnegative polynomials built on the Corteel–Welsh bounded forms; monotonicity of $(H_{c,m})_m$ follows. The positivity of $Q_{n,c}$ at $d = 4$ is Warnaar’s result; the bounded expansion, and two absorption identities that make the two “wall” orbits manifestly positive, appear to be new. The elementary q -binomial infrastructure and the $d = 4$ argument have been formally verified in Lean 4/Mathlib.

1 Introduction

Let $\text{GK}_c(z, q)$ denote the generating function of cylindric partitions of rank 3 and profile $c = (c_0, c_1, c_2)$, in the notation of [7], with z tracking the largest part; set $d := c_0 + c_1 + c_2$ (the level), so the associated modulus is $d + 3$. Since GK_c is invariant under cyclic rotation of the profile, every statement below about a profile applies verbatim to its rotation orbit; we fix once and for all the profile representatives used by Corteel–Welsh [3] at $d = 4$, namely $(2, 1, 1)$, $(4, 0, 0)$, $(3, 1, 0)$, $(3, 0, 1)$, $(2, 2, 0)$, which lie in the five distinct orbits.

Following [7], for $\ell := \gcd(d, 3)$ define

$$Q_{n,c}(q) := (q^\ell; q^\ell)_n [z^n] \left((zq; q)_\infty \text{GK}_c(z, q) \right). \quad (1)$$

The positivity conjecture, stated by Corteel–Dousse–Uncu [2, Conjecture 4.2] and reproduced as [7, Conjecture 2.7], reads:

Conjecture 1.1 ([7, Conjecture 2.7]). *Let $c = (c_0, c_1, c_2)$ and $d := c_0 + c_1 + c_2$ such that $d \not\equiv 0 \pmod{3}$. Then $Q_{n,(c_0,c_1,c_2)}(q)$ is a polynomial in q with nonnegative coefficients. Moreover, $Q_{n,(c_0,c_1,c_2)}(1) = \left(\frac{1}{6}(d+1)(d+2) - 1\right)^n$.*

Polynomiality and the evaluation at $q = 1$ were proved by Welsh [8]; Warnaar proved positivity for $k = 1$ and $k = 2$, i.e. for all profiles with $d \in \{2, 4, 5\}$ [7]; positivity for general d remains open. For the balanced profile $(k, k, k - 1)$, $d = 3k - 1$, Warnaar conjectured an explicit fermionic form:

Conjecture 1.2 ([7, Conjecture 2.5], first identity). *For a positive integer k ,*

$$\text{GK}_{(k,k,k-1)}(z, q) = \frac{1}{(zq; q)_\infty} \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{n_k^2}}{(q; q)_{n_1}} \prod_{i=1}^{k-1} q^{n_i^2 - n_i m_i + m_i^2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix},$$

where $m_k := 2n_k$.

The modulus- $(3k + 1)$ companion concerns the balanced profile $(k, k - 1, k - 1)$ at level $d = 3k - 2$:

Conjecture 1.3 ([7, Conjecture 2.11], first identity). *For a positive integer k ,*

$$\text{GK}_{(k,k-1,k-1)}(z, q) = \frac{1}{(zq; q)_\infty} \sum_{\substack{n_1, \dots, n_{k-1} \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{\sum_{i=1}^{k-1} (n_i^2 - n_i m_i + m_i^2)}}{(q; q)_{n_1}} \times \begin{bmatrix} 2n_{k-1} \\ m_{k-1} \end{bmatrix} \prod_{i=1}^{k-2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix}.$$

Warnaar proved Conjecture 1.2 for $k = 1$ and $k = 2$ [7, Theorem 2.6]; the $k = 1$ case of Conjecture 1.3 is trivial and $k = 2$ is due to Corteel–Welsh [3, Theorem 3.2]. As Warnaar observes, either conjecture at general k immediately implies manifestly positive representations for the $Q_{n,c}$ of its balanced profile, hence Conjecture 1.1 there.

The purpose of this note is threefold.

First (Section 2), we record a simple but, we believe, clarifying observation: for $\gcd(d, 3) = 1$, writing $F_{c,m}(q)$ for the generating function of cylindric partitions of profile c with largest part at most m , and $H_{c,m} := (q; q)_m F_{c,m}$, the family $(H_{c,m})_m$ has a *unique* expansion $H_{c,m} = \sum_{n \leq m} \begin{bmatrix} m \\ n \end{bmatrix} a_n$ with a_n independent of m , and necessarily $a_n = Q_{n,c}$. Consequently Conjecture 1.1 at c is *equivalent* to the existence of a coefficientwise-nonnegative q -binomial expansion of $(H_{c,m})_m$. This is a precise sense in which the problem of bounded A_2 Andrews–Gordon identities and the positivity conjecture are two faces of one problem.

Second (Section 3), we prove Conjecture 1.2 for $k = 3$ ($d = 8$, modulus 11) and Conjecture 1.3 for $k = 4$ ($d = 10$, modulus 13). Each proof is a short chain of results from the literature: Warnaar’s finite form [7, Proposition 7.1] in a coefficientwise limit, an exact single-step Pochhammer split, and one of Uncu’s theorems [6, Theorems 4.3 and 5.3]; the modulus-11 chain additionally needs one contiguous relation of Kanade–Russell [4, Lemma 9.2], while the modulus-13 chain — although it concerns the larger k — is strictly shorter and needs none. We make no claim of depth here — all the hard analysis is in [7, 4, 6] — but the resulting statements appear to be new, and they settle the first cases of Conjecture 1.1 at $d = 8$ and $d = 10$.

Third (Section 4), we revisit $d = 4$, where positivity is Warnaar’s result. Starting from the bounded generating functions of Corteel–Welsh [3, Theorem 1.2] we obtain, for all five orbits including the two “wall” orbits $(3, 0, 1)$ and $(2, 2, 0)$, the bounded fermionic-form expansion

$H_{c,m} = \sum_{n \leq m} \begin{bmatrix} m \\ n \end{bmatrix} Q_{n,c}$ together with explicit manifestly nonnegative formulas for the $Q_{n,c}$. The wall orbits require two absorption identities (Lemmas 4.1 and 4.2), proved by elementary q -Pascal manipulations; Lemma 4.2 is an exact identity, not merely an inequality. Monotonicity $H_{c,m} \geq H_{c,m-1}$ follows for all orbits.

Section 5 describes a formal verification: the q -binomial infrastructure of Section 2, both absorption lemmas, the positivity of all five $d = 4$ forms, and the assembly logic of Section 3 have been proved in Lean 4 over Mathlib, with the literature inputs entering as explicitly named hypotheses. We also report exact $\mathbb{Z}[q]$ verification of Conjecture 1.1 across fourteen levels up to $d = 31$.

Finally, Appendix A describes how this note was produced: the mathematics was generated and checked by an orchestrated system of AI agents, with the author acting as guarantor of the results.

2 The q -binomial expansion of the bounded family

Throughout this section fix a profile c with $\ell = \gcd(d, 3) = 1$, and work in $\mathbb{Z}[[q]]$. Let g_m be the generating function of cylindric partitions of profile c whose largest part is exactly m (so $g_0 = 1$), let $F_{c,m} = \sum_{j \leq m} g_j$, and set

$$H_m := (q; q)_m F_{c,m}, \quad Q_n := Q_{n,c}(q) = (q; q)_n [z^n] \left((zq; q)_\infty \sum_{m \geq 0} g_m z^m \right).$$

Lemma 2.1 (truncated Euler telescoping). *For every integer $K \geq 0$,*

$$\sum_{k=0}^K \frac{(-1)^k q^{\binom{k}{2}}}{(q; q)_k} = \frac{(-1)^K q^{\binom{K+1}{2}}}{(q; q)_K}.$$

Proof. Induction on K : both sides are 1 at $K = 0$, and adding the $(K+1)$ -st term to the right side gives $(-1)^{K+1} q^{\binom{K+1}{2}} (1 - (1 - q^{K+1})) / (q; q)_{K+1} = (-1)^{K+1} q^{\binom{K+2}{2}} / (q; q)_{K+1}$. $\square$

Theorem 2.2 (the Q -transform). *For every $n \geq 0$, $Q_n = \sum_{m=0}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} H_m$.*

Proof. By Euler's expansion $(x; q)_\infty = \sum_{k \geq 0} (-1)^k q^{\binom{k}{2}} x^k / (q; q)_k$ at $x = zq$, extracting the coefficient of z^n ,

$$Q_n = (q; q)_n \sum_{m=0}^n \frac{(-1)^{n-m} q^{\binom{n-m}{2}}}{(q; q)_{n-m}} g_m. \quad (2)$$

Substituting $H_m = (q; q)_m \sum_{j \leq m} g_j$ into the claimed sum, using $\begin{bmatrix} n \\ m \end{bmatrix} (q; q)_m = (q; q)_n / (q; q)_{n-m}$, interchanging the (finite) sums, and applying Lemma 2.1 to the inner sum over $k = n - m$ reproduces (2). $\square$

Theorem 2.3 (Gauss inversion; classical). *For sequences $(a_n), (b_n)$ in $\mathbb{Z}[[q]]$ the statements (i) $b_n = \sum_{j \leq n} \begin{bmatrix} n \\ j \end{bmatrix} a_j$ for all n and (ii) $a_n = \sum_{m \leq n} (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} b_m$ for all n are equivalent; moreover for given (b_n) the sequence (a_n) in (i) is unique.*

Proof. Both transforms are lower-unitriangular, so it suffices to check the orthogonality

$$\sum_{m=k}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} \begin{bmatrix} m \\ k \end{bmatrix} = \delta_{n,k}.$$

Using $\begin{bmatrix} n \\ m \end{bmatrix} \begin{bmatrix} m \\ k \end{bmatrix} = \begin{bmatrix} n \\ k \end{bmatrix} \begin{bmatrix} n-k \\ m-k \end{bmatrix}$ and substituting $i = n - m$, $J = n - k$, the left side equals

$$\begin{bmatrix} n \\ k \end{bmatrix} \sum_{i=0}^J (-1)^i q^{\binom{i}{2}} \begin{bmatrix} J \\ i \end{bmatrix} = \begin{bmatrix} n \\ k \end{bmatrix} (x; q)_J \Big|_{x=1} = \begin{bmatrix} n \\ k \end{bmatrix} \delta_{J,0}$$

by Gauss's finite q -binomial theorem. $\square$

Corollary 2.4 (inverse Q -transform; unconditional). *For every $m \geq 0$, $H_m = \sum_{n=0}^m \begin{bmatrix} m \\ n \end{bmatrix} Q_n$.*

At $q = 1$ this reconciles Welsh's evaluation with the orbit count: $H_m(1) = \sum_n \binom{m}{n} (K-1)^n = K^m$, $K = \frac{1}{6}(d+1)(d+2)$.

Proposition 2.5 (equivalence). *Fix c with $d \not\equiv 0 \pmod{3}$.*

- (1) *If $H_m = \sum_{j \leq m} \begin{bmatrix} m \\ j \end{bmatrix} a_j$ for all m with $a_j \in \mathbb{Z}[[q]]$ independent of m , then $a_j = Q_j$ for all j . In particular, any bounded fermionic form for $(H_m)_m$ — an expansion of this shape with manifestly nonnegative a_j — proves Conjecture 1.1 at c , with the explicit formula $Q_n = a_n$.*
- (2) *The family $(H_m)_m$ admits a coefficientwise-nonnegative q -binomial expansion if and only if Conjecture 1.1 holds at c .*

Proof. (1) is the uniqueness in Theorem 2.3 combined with Corollary 2.4. (2): the unique expansion coefficients are the Q_n , so a nonnegative expansion exists iff they are all nonnegative. $\square$

Thus, for $\gcd(d, 3) = 1$, there is no shortcut criterion certifying the existence of a nonnegative expansion without exhibiting its coefficients: for the family $(H_m)_m$, existence *is* the conjecture.

3 Conjectures 1.2 and 1.3 beyond $k = 2$: the profiles $(3, 3, 2)$ and $(4, 3, 3)$

Write $\text{FERM}_k(z, q)$ for the multisum in Conjecture 1.2 (without the $1/(zq; q)_\infty$ prefactor), and $G_c(z, q) := (zq; q)_\infty \text{GK}_c(z, q)$, $\mathcal{H}_c(z, q) := G_c(z, q)/(q; q)_\infty$. For $\rho, \sigma \in \mathbb{Z}^3$ let, following Uncu [6, Eq. (2.8)] (the case of modulus 11), abbreviating

$$P_{r,s} := (q; q)_{r_1-r_2} (q; q)_{r_2-r_3} (q; q)_{r_3} (q; q)_{s_1-s_2} (q; q)_{s_2-s_3} (q; q)_{s_3},$$

$$S_{11}(z; \rho | \sigma) := \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2 + \rho_i r_i + \sigma_i s_i) + 2r_3 s_3}}{P_{r,s} (q; q)_{r_3+s_3+1}}, \quad (3)$$

and abbreviate $e_3 = (0, 0, 0)$, $e_2 = (0, 0, 1)$, $e_1 = (0, 1, 1)$, with δ_i the i -th unit vector. We suppress z and write $S(\rho | \sigma)$.

Theorem 3.1. $G_{(3,3,2)}(z, q) = \text{FERM}_3(z, q)$; that is, Conjecture 1.2 holds for $k = 3$.

Corollary 3.2. For all $n \geq 0$,

$$Q_{n,(3,3,2)}(q) = \sum_{\substack{n \geq n_2 \geq n_3 \geq 0 \\ m_1, m_2 \geq 0}} q^{n^2 + n_2^2 + n_3^2 - n m_1 - n_2 m_2 + m_1^2 + m_2^2} \begin{bmatrix} n \\ n_2 \end{bmatrix} \begin{bmatrix} n - n_2 + m_2 \\ m_1 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \begin{bmatrix} n_2 + n_3 \\ m_2 \end{bmatrix},$$

a manifestly nonnegative polynomial. In particular Conjecture 1.1 holds for the orbit of $(3, 3, 2)$ at $d = 8$.

Proof of the corollary. $Q_{n,(3,3,2)} = (q; q)_n [z^n] \text{FERM}_3$; the summand at $n_1 = n$ carries the denominator $(q; q)_n$, which cancels. (Here $n_2 - n_3 + m_3 = n_2 + n_3$ since $m_3 = 2n_3$, and the exponent n_3^2 has been absorbed into the displayed quadratic form.) $\square$

Proof of Theorem 3.1. The proof chains four proved statements.

Step 1: Warnaar's finite form in the limit. For $\sigma = (1, \dots, 1, a)$, $a \in \{-1, 1\}$, Warnaar defines [7, Eqs. (7.1), (7.2)]

$$F_{n_0, m_0; k}^{(a)}(z, q) = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_k \geq 0}} \frac{z^{n_1}}{(q; q)_{n_k + m_k}} \prod_{i=1}^k q^{n_i^2 - \sigma_i n_i m_i + m_i^2} \begin{bmatrix} n_{i-1} \\ n_i \end{bmatrix} \begin{bmatrix} m_{i-1} \\ m_i \end{bmatrix},$$

and proves [7, Proposition 7.1, case $a = -1$] the polynomial identity

$$F_{n_0, m_0; k}^{(-1)}(z, q) = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{n_k^2}}{(q; q)_{m_0 - n_1 + m_1}} \begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \prod_{i=1}^{k-1} q^{n_i^2 - n_i m_i + m_i^2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix}, \quad (4)$$

where $m_k := 2n_k$. Fix $k = 3$ and let $n_0, m_0 \rightarrow \infty$. Each coefficient of $z^n q^N$ on both sides stabilizes; the right side of (4) becomes $\text{FERM}_3(z, q)/(q; q)_\infty$, while on the left the two binomial chains telescope, $\begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \rightarrow 1/((q; q)_{n_1 - n_2} (q; q)_{n_2 - n_3} (q; q)_{n_3})$ and likewise for the m -chain. Renaming $(n, m) \rightarrow (r, s)$ and using $\sigma = (1, 1, -1)$, so that the $i = 3$ exponent is $r_3^2 + r_3 s_3 + s_3^2 = (r_3^2 - r_3 s_3 + s_3^2) + 2r_3 s_3$, we obtain

$$\frac{\text{FERM}_3(z, q)}{(q; q)_\infty} = T(z, q) := \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2) + 2r_3 s_3}}{P_{r, s}(q; q)_{r_3 + s_3}}. \quad (5)$$

Step 2: Pochhammer split. Using $1/(q; q)_{r_3 + s_3} = (1 - q^{r_3 + s_3 + 1})/(q; q)_{r_3 + s_3 + 1}$ termwise, together with $q^{r_3 + s_3 + 1} = q \cdot q^{r_3} \cdot q^{s_3}$ — which realizes the subtracted term as the shift $(\rho_3, \sigma_3) \mapsto (\rho_3 + 1, \sigma_3 + 1)$ in (3) with a global factor q — we get

$$T(z, q) = S(e_3 | e_3) - q S(e_2 | e_2). \quad (6)$$

Step 3: one Kanade–Russell contiguous relation. Kanade–Russell [4, Lemma 9.2] prove, for modulus $\equiv -1 \pmod{3}$ and $\sigma_{k-1} = 0$ (in the notation (3), $\sigma_2 = 0$), the relation

$$S(\rho | \sigma) - S(\rho | \sigma + \delta_3) - q S(\rho + \delta_3 | \sigma + \delta_3) + q S(\rho + \delta_3 | \sigma + \delta_2 + \delta_3) = 0.$$

At $\rho = \sigma = (0, 0, 0)$ this reads $S(e_3|e_3) - S(e_3|e_2) - qS(e_2|e_2) + qS(e_2|e_1) = 0$, i.e.

$$S(e_3|e_3) - qS(e_2|e_2) = S(e_3|e_2) - qS(e_2|e_1). \quad (7)$$

Step 4: Uncu's modulus-11 theorem. Uncu [6, Eq. (4.1) and Theorem 4.3] proves

$$\mathcal{H}_{(3,3,2)}(z, q) = S(e_3|e_2) - qS(e_2|e_1). \quad (8)$$

Combining (5)–(8), $\text{FERM}_3(z, q) = (q; q)_\infty \mathcal{H}_{(3,3,2)}(z, q) = G_{(3,3,2)}(z, q)$. $\square$

The profile (4, 3, 3) at modulus 13

Write $\text{FERM}_{k-1}^+(z, q)$ for the multisum in Conjecture 1.3 (without the $1/(zq; q)_\infty$ prefactor). Following Uncu [6, Eq. (2.10)] (the case of modulus 13), define

$$S_{13}(z; \rho | \sigma) := \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2 + \rho_i r_i + \sigma_i s_i)}}{P_{r,s}(q; q)_{r_3 + s_3 + 1}}; \quad (9)$$

note that, in contrast to (3), there is *no* cross term $q^{2r_3 s_3}$ in the modulus-13 family. We again suppress z .

Theorem 3.3. $G_{(4,3,3)}(z, q) = \text{FERM}_3^+(z, q)$; that is, Conjecture 1.3 holds for $k = 4$.

Corollary 3.4. For all $n \geq 0$,

$$Q_{n,(4,3,3)}(q) = \sum_{\substack{n_2, n_3 \geq 0 \\ m_1, m_2, m_3 \geq 0}} q^{n^2 + n_2^2 + n_3^2 - nm_1 - n_2 m_2 - n_3 m_3 + m_1^2 + m_2^2 + m_3^2} \\ \times \begin{bmatrix} n \\ n_2 \end{bmatrix} \begin{bmatrix} n - n_2 + m_2 \\ m_1 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \begin{bmatrix} n_2 - n_3 + m_3 \\ m_2 \end{bmatrix} \begin{bmatrix} 2n_3 \\ m_3 \end{bmatrix},$$

a manifestly nonnegative polynomial. In particular Conjecture 1.1 holds for the orbit of (4, 3, 3) at $d = 10$.

Proof of Theorem 3.3 (sketch). The chain has three links — Steps 1, 2 and 4 above have exact analogues, and there is no analogue of Step 3. A detailed writeup is in the companion note [5].

Step 1'. The $a = +1$ case of [7, Proposition 7.1] at $k = 3$ (all $\sigma_i = 1$), in the same coefficientwise limit $n_0, m_0 \rightarrow \infty$ as in Step 1, gives

$$\frac{\text{FERM}_3^+(z, q)}{(q; q)_\infty} = \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2)}}{P_{r,s}(q; q)_{r_3 + s_3}} =: T'(z, q),$$

with no cross term, since $\sigma_3 = +1$ leaves the quadratic form $r_3^2 - r_3 s_3 + s_3^2$ intact.

Step 2'. The identical Pochhammer split $1/(q; q)_{r_3 + s_3} = (1 - q^{r_3 + s_3 + 1})/(q; q)_{r_3 + s_3 + 1}$ gives, termwise, $T' = S_{13}(e_3|e_3) - qS_{13}(e_2|e_2)$.

Step 4'. Uncu [6, Eq. (5.1), last entry, and Theorem 5.3] proves $\mathcal{H}_{(4,3,3)}(z, q) = S_{13}(e_3|e_3) - qS_{13}(e_2|e_2)$ — *verbatim* the right side of Step 2'. Hence $\text{FERM}_3^+ = (q; q)_\infty \mathcal{H}_{(4,3,3)} = G_{(4,3,3)}$. $\square$

Remark 3.5 (why the $d = 10$ chain is shorter). *At modulus $3k - 1$ the finite form produces the pair $(e_3|e_3), (e_2|e_2)$ while the proved Kanade–Russell expression lists $(e_3|e_2), (e_2|e_1)$, so one contiguous relation is needed to bridge them (Step 3). At modulus $3k + 1$ the Kanade–Russell pattern $\mathcal{H}_{(c_1, c_2, c_3)} = S(e_{c_2}|e_{c_3}) - qS(e_{c_2-1}|e_{c_3-1})$, evaluated at the balanced profile $c_2 = c_3 = k - 1$, is exactly the split pair, and no bridge is needed. Thus the template — finite form, coefficientwise limit, Pochhammer split, Uncu — scales directly to whichever moduli have proved Kanade–Russell expressions, with the contiguous-relation step required only at moduli $\equiv -1 \pmod{3}$.*

Remark 3.6 (provenance of Uncu’s theorems). *Uncu’s Theorems 4.3 and 5.3 [6] are computer-assisted proofs: they rest on ideal-membership computations (Gaussian elimination over q -difference operators) whose certificates are provided in the ancillary files of arXiv:2301.01359. Theorems 3.1 and 3.3 inherit that dependency.*

4 $d = 4$: bounded fermionic forms and the absorption identities

Fix $d = 4$ (modulus 7), and recall the five orbit representatives $(2, 1, 1), (4, 0, 0), (3, 1, 0), (3, 0, 1), (2, 2, 0)$ of Section 1. Write $T(n, j) := q^{n^2 + j^2 - nj}$. Corteel–Welsh [3, Theorem 1.2] give the bounded generating functions

$$F_{c,m} = \sum_{n \leq m} \frac{\tilde{A}_{n,c}}{(q; q)_{m-n}},$$

with m -independent numerators $\tilde{A}_{n,c}$; explicitly, reading off from their theorem (equivalently from the G_c forms of [3, Theorem 3.2]),

$$\begin{aligned} \tilde{A}_n^{(2,1,1)} &= \sum_{j=0}^{2n} \frac{T(n, j)}{(q; q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}, & \tilde{A}_n^{(4,0,0)} &= \sum_j \frac{T(n, j) q^{n+j}}{(q; q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}, \\ \tilde{A}_n^{(3,1,0)} &= \sum_j \frac{T(n, j) q^j}{(q; q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix}, \\ \tilde{A}_n^{(3,0,1)} &= \sum_j T(n, j) \left(\frac{q^n}{(q; q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix} + \frac{q^{2j}}{(q; q)_{n-1}} \begin{bmatrix} 2n-2 \\ j \end{bmatrix} \right), \\ \tilde{A}_n^{(2,2,0)} &= \sum_j T(n, j) \left(\frac{q^n}{(q; q)_n} \begin{bmatrix} 2n \\ j \end{bmatrix} + \frac{q^j(1 + q^{n+j})}{(q; q)_{n-1}} \begin{bmatrix} 2n-2 \\ j \end{bmatrix} \right), \end{aligned}$$

with the convention $1/(q; q)_{-1} = 0$.

Since the $\tilde{A}_{n,c}$ are independent of m , letting $m \rightarrow \infty$ gives $Q_{n,c} = (q; q)_n \tilde{A}_{n,c}$, and Proposition 2.5(1) (or a one-line direct computation with $\begin{bmatrix} m \\ n \end{bmatrix} = (q; q)_m / ((q; q)_{m-n} (q; q)_n)$) yields

$$H_{c,m} = \sum_{n=0}^m \begin{bmatrix} m \\ n \end{bmatrix} Q_{n,c} \quad \text{for all five orbits.} \quad (10)$$

Using $(q; q)_n / (q; q)_{n-1} = 1 - q^n$, the coefficients are

$$\begin{aligned} Q_n^{(2,1,1)} &= \sum_j T(n, j) \begin{bmatrix} 2n \\ j \end{bmatrix}, & Q_n^{(4,0,0)} &= \sum_j T(n, j) q^{n+j} \begin{bmatrix} 2n \\ j \end{bmatrix}, & Q_n^{(3,1,0)} &= \sum_j T(n, j) q^j \begin{bmatrix} 2n \\ j \end{bmatrix}, \\ Q_n^{(3,0,1)} &= X_n + (1 - q^n) X'_n, & Q_n^{(2,2,0)} &= X_n + (1 - q^n) Y'_n, \end{aligned} \quad (11)$$

where

$$\begin{aligned} X_n &= \sum_j T(n, j) q^n \begin{bmatrix} 2n \\ j \end{bmatrix}, & X'_n &= \sum_j T(n, j) q^{2j} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}, \\ Y'_n &= \sum_j T(n, j) q^j (1 + q^{n+j}) \begin{bmatrix} 2n-2 \\ j \end{bmatrix}. \end{aligned}$$

The first three forms are manifestly nonnegative. The two wall orbits (11) are not, because of the factor $1 - q^n$; positivity follows from the absorption inequalities $X_n \geq q^n X'_n$ and $X_n \geq q^n Y'_n$, which we now prove.

Lemma 4.1 (Absorption A). *For all $n \geq 1$,*

$$X_n - q^n X'_n = \sum_j T(n, j) q^n \left((q^{j-1} + q^j) \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix} + \begin{bmatrix} 2n-2 \\ j-2 \end{bmatrix} \right) \geq 0.$$

Proof. Two applications of the q -Pascal rule $\begin{bmatrix} N \\ j \end{bmatrix} = \begin{bmatrix} N-1 \\ j-1 \end{bmatrix} + q^j \begin{bmatrix} N-1 \\ j \end{bmatrix}$ give

$$\begin{bmatrix} 2n \\ j \end{bmatrix} = q^{2j} \begin{bmatrix} 2n-2 \\ j \end{bmatrix} + (q^{j-1} + q^j) \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix} + \begin{bmatrix} 2n-2 \\ j-2 \end{bmatrix}.$$

The first term contributes exactly q^n times the j -th summand of X'_n ; the remainder is nonnegative. $\square$

Lemma 4.2 (Absorption B; exact). *For all $n \geq 1$,*

$$X_n - q^n Y'_n = \sum_{j=0}^{2n-1} q^{n^2+j^2-nj+2j+1} \begin{bmatrix} 2n-1 \\ j \end{bmatrix} \geq 0.$$

Proof. Divide by the global factor q^{n^2+n} and set

$$D_n = \sum_j q^{j^2-nj} \begin{bmatrix} 2n \\ j \end{bmatrix} - \sum_j q^{j^2-nj} (q^j + q^{n+2j}) \begin{bmatrix} 2n-2 \\ j \end{bmatrix}.$$

Step 1. The two q -Pascal rules $\begin{bmatrix} 2n \\ j \end{bmatrix} = \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix} + q^j \begin{bmatrix} 2n-1 \\ j \end{bmatrix}$ and $\begin{bmatrix} 2n-1 \\ j \end{bmatrix} = \begin{bmatrix} 2n-2 \\ j \end{bmatrix} + q^{2n-1-j} \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}$ combine to

$$\begin{bmatrix} 2n \\ j \end{bmatrix} - q^j \begin{bmatrix} 2n-2 \\ j \end{bmatrix} = \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix} + q^{2n-1} \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix}, \quad (12)$$

valid for all $0 \leq j \leq 2n$ with $\begin{bmatrix} N \\ k \end{bmatrix} := 0$ for $k < 0$ or $k > N$.

Step 2. Substituting (12),

$$D_n = \sum_j q^{j^2-nj} \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix} + \sum_j q^{j^2-nj+2n-1} \begin{bmatrix} 2n-2 \\ j-1 \end{bmatrix} - \sum_{j=0}^{2n-2} q^{j^2-nj+n+2j} \begin{bmatrix} 2n-2 \\ j \end{bmatrix}.$$

In the last sum replace $j \mapsto j-1$: the exponent becomes $(j-1)^2 - n(j-1) + n + 2(j-1) = j^2 - nj + 2n - 1$, so it cancels the middle sum *exactly*. Hence $D_n = \sum_{j \geq 1} q^{j^2-nj} \begin{bmatrix} 2n-1 \\ j-1 \end{bmatrix}$; re-indexing $j \mapsto j+1$ and restoring q^{n^2+n} gives the claim. $\square$

Theorem 4.3. *For every profile c with $c_0 + c_1 + c_2 = 4$:*

- (i) $Q_{n,c}(q) \geq 0$ coefficientwise for all n , with the manifestly positive formulas above (walls via Lemmas 4.1, 4.2); $Q_{n,c}(1) = 4^n$.
- (ii) (bounded fermionic form at $d = 4$) $H_{c,m} = \sum_{n \leq m} \begin{bmatrix} m \\ n \end{bmatrix} Q_{n,c}$ with $Q_{n,c} \geq 0$, for all five orbits including the walls.
- (iii) (monotonicity) $H_{c,m} - H_{c,m-1} = \sum_{n \geq 1} q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix} Q_{n,c} \geq 0$.

Proof. (i): the forms (11) plus Lemmas 4.1 and 4.2. (ii): equation (10). (iii): the q -Pascal rule $\begin{bmatrix} m \\ n \end{bmatrix} - \begin{bmatrix} m-1 \\ n \end{bmatrix} = q^{m-n} \begin{bmatrix} m-1 \\ n-1 \end{bmatrix}$ applied termwise to (ii). $\square$

We emphasize that part (i) is not new: positivity of $Q_{n,c}$ for all profiles with $d \in \{2, 4, 5\}$ is Warnaar’s result [7]. What we believe to be worth recording is the bounded refinement (ii)–(iii) — by Proposition 2.5 the expansion (10) is the *unique* bounded fermionic form at the first level — and the absorption mechanism itself: the wall orbits satisfy $Q_n = P_n + (1 - q^n)P'_n$ with $P_n, P'_n \geq 0$, and positivity reduces to an inequality $P_n \geq q^n P'_n$ provable purely by q -Pascal manipulation (a double-Pascal split for one wall, an exact shift-cancellation for the other). It seems plausible that this pattern persists for the non-balanced orbits at higher d , where positive systems of Corteel–Welsh type are not yet available.

5 Machine verification

The elementary layers of Sections 2–4 have been formalized in Lean 4 over Mathlib; the development (library `WarnaarGlue`, five modules, no `sorry`) is available at

<https://github.com/RaggedR/warnaar-glue>

q -binomial coefficients are defined by the Pascal recursion over an arbitrary commutative ring (specialized to $\mathbb{Z}[q] = \text{Polynomial } \mathbb{Z}$ where positivity is involved), and nonnegativity is the predicate “every coefficient of the polynomial is ≥ 0 ”. Table 1 maps the statements of this note to the formal theorems.

The division of labour is made explicit in the formalization: results proved here from scratch (the q -binomial identities, both absorption lemmas, positivity of the five forms, the inversion) are unconditional Lean theorems, while the literature inputs enter as *named hypotheses* of the main theorems — `hCW` (the Corteel–Welsh bounded forms [3, Theorem 1.2], with denominators cleared), `hQ` (the Q -transform relation of Theorem 2.2, whose Euler-series proof is not formalized), and, for the $d = 8$ chain, `hWarnaar` (Step 1), `hSplit` (the series form of Step 2, whose polynomial kernel is formalized as `qpoch.split`), `hBridge` (Step 3, [4, Lemma 9.2]) and `hUncu` (Step 4, [6, Theorem 4.3]). The $d = 10$ chain of Theorem 3.3 has *not* been formalized; its hypotheses would be the exact analogues of the $d = 8$ ones (with no bridge). A machine-checked non-vacuity witness shows that the hypotheses of the $d = 4$ theorems are satisfiable. Every theorem in the library depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

Separately from the formal proofs, Conjecture 1.1 was verified in exact $\mathbb{Z}[q]$ arithmetic (no series truncation), for *every* profile at each level, at

$$d \in \{2, 4, 5, 7, 8, 13, 16, 17, 19, 20, 22, 23, 25, 31\},$$

Statement in this note	Lean theorem(s)
q -Pascal rules	<code>pascal₁</code> , <code>pascal₂</code>
Gauss’s finite q -binomial theorem at $x = 1$	<code>alt_sum</code>
orthogonality in Theorem 2.3	<code>orth_ML</code> , <code>orth_LM</code>
Theorem 2.3 (Gauss inversion, both directions)	<code>qbinom_inversion</code>
Theorem 2.2 $\leftrightarrow$ Corollary 2.4 (abstract form)	<code>Q_transform_of_H</code> , <code>H_of_Q_transform</code>
monotonicity kernel (Theorem 4.3(iii))	<code>pascal_ladder</code>
Pochhammer split kernel of Step 2, Section 3	<code>qpoch_split</code> , <code>qpoch_split_shift</code>
assembly of Steps 1–4 ($d = 8$ chain)	<code>seed2_assembly</code> , <code>seed2_chain</code>
$d = 10$ chain (Theorem 3.3)	not formalized (cf. <code>seed2_chain</code>)
Lemma 4.1	<code>absorption_A</code>
Lemma 4.2	<code>absorption_B</code>
positivity of all five $d = 4$ forms	<code>Qcw_nonneg</code>
identification $Q_{n,c} = (q; q)_n \tilde{A}_{n,c}$ via inversion	<code>d4_Q_eq_Qcw</code>
Theorem 4.3(i)	<code>d4_positive</code>
Theorem 4.3(ii)	<code>d4_BFF</code>
Theorem 4.3(iii)	<code>d4_monotone</code>

Table 1: Paper-to-Lean dictionary.

in the ranges $n \leq 16$ for $d = 13$; $n \leq 12$ for $d = 16, 17, 19, 20$; $n \leq 10$ for $d = 22, 23$; $n \leq 9$ for $d = 25$; $n \leq 7$ for $d = 31$; and in extended ranges at small level ($n \leq 25$ at $d = 4$, $n \leq 22$ at $d = 5$, $n \leq 18$ at $d = 7$, $n \leq 16$ at $d = 8$, $n \leq 12$ at $d = 13$, wall orbits included). The minimal coefficients show no shrinking-margin trend as d grows.

Acknowledgements

Everything here is built on the work of Corteel–Welsh, Corteel–Dousse–Uncu, Kanade–Russell, Uncu, Welsh and Warnaar; the contribution of this note is only to observe that their results, chained together, prove a little more than has so far been recorded.

A Methodology

The results of this note were produced by an orchestrated system of AI agents (Claude, Anthropic), and the author believes it is right to say so plainly. This appendix records how the system worked and what was done to keep it honest. The author has read the mathematics of this note in full and vouches for it; the division of labour was that the AI system generated and checked the proofs, while the human role was that of guarantor.

Corpus and seeding

The system worked over a retrieval corpus of 82 papers on Rogers–Ramanujan-type identities, cylindric partitions and neighbouring topics, split into roughly 7,000 chunks and embedded with a fine-tuned SPECTER2 model. Eight initial research directions were selected by k -medoids clustering of the embedded corpus, so that the starting points were spread across genuinely different regions of the literature (bounded forms, contiguous relations, vertex operators, bijective methods, and so on) rather than chosen by hand.

The loop

Each round (“layer”) ran eight agents in parallel, one per research direction, each attempting proof strategies and recording partial progress, failures and computations in a working file. A ninth agent then synthesized the eight working files into a single document, and the next layer was re-seeded from that synthesis. Later rounds allowed the agents to generate their own retrieval queries against the corpus as their attempts developed. The chain of results in this note — the Q -transform equivalence, the four-step $d = 8$ assembly, and the $d = 4$ absorption lemmas — emerged over several such layers.

Discipline

Since language models can produce confident errors, the system was built around redundancy and adversarial checking rather than trust:

- One seed was a dedicated *adversary*, tasked not with proving anything but with independently recomputing others’ claims and hunting for counterexamples.
- Any claim of a completed proof triggered independent verifier agents, which re-derived the argument from the primary sources without access to the claimant’s reasoning.
- The load-bearing elementary identities were formalized in Lean 4/Mathlib (Section 5; the repository <https://github.com/RaggedR/warnaar-glue> is public), with the literature inputs isolated as named hypotheses so that the boundary between what is machine-checked and what is cited is explicit.
- All positivity claims were tested by exact $\mathbb{Z}[q]$ computation against an engine that had itself been validated against the raw combinatorial definition of cylindric partitions, in the ranges reported in Section 5.

This redundancy earned its keep: two convention errors (a mislabelling of profile orbits and an orientation error in a contiguous relation) were caught precisely because multiple agents re-derived the same objects independently and their answers disagreed; both were traced and fixed against the primary sources. The standing rule that any bridging contiguous relation be re-derived directly from [4, Lemma 9.2], rather than taken from a restatement, is a lesson from one of these episodes.

What this does and does not claim

No claim is made that the system displayed insight beyond assembling, checking and slightly extending results already in the literature; as stated in the acknowledgements, the hard analysis is due to the cited authors. The point of recording the methodology is transparency: the reader should know that the proofs were found and verified this way, and should weigh the machine-checked and exactly-computed components (Section 5) accordingly.

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